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ABSTRACT

This research analyzes the impact of CSR on the debt maturity structure of banks listed companies in Vietnam. Vietnam's financial market is still new, there are not many regulations to protect investors, resulting in much information asymmetry. In this context, enterprises implementation of CSR is expected to help increase investors' understanding of firms' CSR activities and help businesses issue debt to serve production and business activities. We used two regression methods (OLS and System GMM with instrumental variables) for the sample of 220 companies listed on Vietnam exchange in the period 2014 – 2018. CSR performance is positively correlated with short-term debt, according to our findings. Based on the findings, we suggest solutions for enterprises and stakeholders to actively implement social responsibility to optimize debt financing and integrate into the world economy's development trend.

Keywords: Information asymmetry, debt maturity, long-term debt, short-term debt, corporate social responsibility.

CHAPTER 1: INTRODUCTION

1. THE NECESSITY OF THESIS

Nowadays, the implementation of CSR is being developed, widely distributed, and is gradually becoming an obligatory norm for the majority of multinational companies and enterprises in the world, especially the growing trend of worldwide economic integration. However, CSR is still a new concept that has been introduced into the Vietnamese market for more than a decade. In order to achieve the goal of developing competitiveness and creating more opportunities to penetrate new economic markets with domestic and international organizations, and enterprises in Vietnam, they need to increase and strengthen the implementation of social responsibility.

Recognizing that CSR is a requirement in the integration process of the companies, there have been researched papers on the influence and importance of CSR to businesses in terms of its benefits, its impact on customers, and the reality of CSR information disclosure in Vietnam, etc. Typically "Relationship between social responsibility, business advantages and financial performance of enterprises, Can Tho city area" (Chau Thi Le Duyen, Nguyen Huynh Kim Ngan, and Nguyen Thanh Liem, 2014). Although the above study has obviously shown that utilizing corporate social responsibility will contribute to enhanced financial performance for a long time, there are still some limitations when it comes to the quantity of small data collected from firms, so generalization is restricted, and the research has not surely proved the effectiveness of corporate social responsibility in terms of financial performance.

In addition, for developed countries in the world, the optimization of debt maturity structure and factors affecting debt maturity is a common and attractive issue. Debt maturity structure plays an important role in maximizing the value of firms, and research shows that CSR has an impact on many corporate policies, including sponsorship policy. Furthermore, Nguyen, Choi, & Agbola (2020) suggested that facilitating CSR financing of long-term debt will help shape future policies to support enterprises' participation in CSR. In the current increasingly fierce market competition, the implementation of CSR is still a new activity for many businesses in our country.

Therefore, analyzing and comprehending the influence of CSR on the debt term structure - to select the most appropriate debt capital source is a critical issue that firms

in Vietnam should approach as soon as possible. Moreover, in comparison to developed economies, Vietnam has not experienced comprehensive development in terms of a transition economy, constrained access to financial markets, and numerous information asymmetry issues between internal and external entities, preventing businesses from accessing secured loans. Therefore, our research will help analyze the impacts of CSR on debt maturity structure, in order to offer effective solutions to support Vietnamese enterprises, can further improve operational efficiency in business, solve the problem of information asymmetry, and establish a solid position in the domestic market as well as the global market are extremely necessary.

For the mentioned reasons, with the aim of investigating the impact of CSR on debt maturity structure, our research team decided to choose the topic: **"The impact of CSR on corporate debt maturity structure in Vietnam"** as a research thesis.

2. RESEARCH OBJECTIVES

2.1 Common goals

Analyze the effect of social responsibility implementation on the debt maturity structure of businesses in Vietnam, thereby providing managers, investors, and policymakers with relevant advice and assistance in making optimal financial decisions to boost productivity and improve the ability of the business to operate effectively.

2.2 Specific goals

To achieve common goals, we need to complete two specific objectives:

(1) Design, build a model, and identify the impact of CSR on the term structure debts owed by Vietnamese enterprises;

(2) Proposing, orienting, and giving solutions for companies to be conscious of CSR in debt finance mobilization, assisting firms in increasing their competitive chance and sustainable development.

3. RESEARCH OBJECTS AND SCOPE

3.1 Research object

Research object: CSR and debt maturity structure of enterprises in Vietnam.

3.2 Research scope

- Scope of content: Impact of CSR on the debt maturity structure of enterprises in Vietnam.

- Spatial scope: The research was carried out within enterprises in the territory of Vietnam, with a number of 220 chosen enterprises.

- Scope of time: The research was conducted based on the annual reports and sustainable development in the period from 2014 to 2018 of listed companies in Vietnam.

4. NEW RESEARCH CONTRIBUTION

Theoretically, the research confirms the validity and is consistent with several theories such as shareholder theory, stakeholder theory, signal theory, and especially agency theory. Agency theory suggests that in the relationship between shareholders and principals or managers there is always a conflict of interest between them. Accordingly, instead of investing in potential projects to enhance financial performance, managers may utilize the business's resources to expand their wealth and utility. This leads to the most aggressive application of CSR implementation from managers at the enterprise, moreover, to limit the issuance of loans since creditors will have complete control and dominance over them, and they will not be able to seek personal gains.

From a practical perspective, based on the actual situation in the Vietnamese market with the combination of agency theory, it can be seen that the information asymmetry in Vietnam is significantly prevalent. In detail, objects inside the business hold lots of important information about the company's business operations, and the flow of money,... while outside parties such as banks and investors know less information, and sometimes the information obtained is not accurate. Furthermore, in comparison to developed nations, our country has not created any conditions for the development of investor protection systems, requiring them to be extremely cautious before pouring money into projects and investments to avoid future losses. Because of that, when businesses strengthen the implementation of CSR as an effective measure to significantly reduce the phenomenon of information asymmetry, mobilize external capital to serve their activities, business activities, and have a positive impact on the

value and image of the company in the competitive market, creating conditions for not only businesses but also society to have sustainable development.

5. STRUCTURE OF RESEARCH

Chapter 1: Introduction

Chapter 2: Theoretical framework

Chapter 3: Methodology

Chapter 4: Research results

Chapter 5: Conclusion

CHAPTER 2: THEORETICAL FRAMEWORK

1. OVERVIEW OF CSR

Nowadays, CSR is defined differently depending on the perspective and business model of each enterprise and country. Bowen (1953) argued that "CSR refers to the obligation of a business to pursue policies, make decisions, or perform activities to achieve set goals and social values".

With Maignan, Ferrell (2004) also proposed the definition of CSR as follows: “A business has social responsibility when its decisions and activities aim to create and balance the various benefits of individuals and stakeholders.”. The concept introduced by the World Business Council for sustainable development (World Business Council for sustainable development) of the World Bank (2002), led by Nigel Twose, has been widely used: “Social responsibility is an enterprise's commitment to contribute to sustainable economic development, through activities aimed at improving the quality of life of employees and their families; community and society, in a way that benefits both businesses and society as a whole.

Benefits of implementing CSR:

(1) Reduce costs and improve productivity,

(2) Increase profits,

(3) Increase the brand value and reputation of the business,

(4) Attracting quality labor resources,

(5) Create opportunities to access (enter) new markets,

(6) Exploiting opportunities from implementing social responsibility,

(7) Employee and customer loyalty (Nguyen Dinh Cung and Luu Duc Minh, 2008).

2. LITERATURE REVIEW

2.1 Shareholder Theory

Derived from the theories of the company with the idea that the primary purpose of the corporation is to maximize profit value. Following this initial viewpoint, Friedman (1962, 1970) claimed that firms should focus solely on profit maximization and should not be required to engage in CSR. In other words, the most important people are shareholders and the company must use all its resources to increase shareholder

wealth. As a result, CSR implementation has been limited and ineffective. On the other hand, according to pecking order theory, firstly, the company will take advantage of retained earnings to maintain the expansion of production and business activities while ignoring the narrowing of the gap to long-term loans. Therefore, we can expect that CSR has a negative relationship with debt maturity structure.

2.2 Agency Theory

According to agency theory, there is always a conflict of interest between shareholders and business managers. This conflict is often referred to as the "representation problem". The point of this theory is that it is difficult to reconcile interests between shareholders and managers of the firm and that conflicts can result from social responsibility (Jensen and Meckling, 1976). Accordingly, managers can utilize the resources of the business to increase their own wealth and utility instead of investing in potential projects so that financial performance can be improved. As a result, socially responsible activities can lead to poor performance of the company and reduce shareholders' assets. This leads to the maximum active implementation of CSR from managers at the enterprise, moreover, to limit the issuance of loans because then they will be under the control of creditors, with no opportunity to seek their own personal interests. According to the results, the research team expects that CSR has a negative relationship with debt maturity structure.

2.3 Stakeholder theory

The stakeholder perspective from Freeman (1984) suggests that a business can only survive if it can meet the requirements of its stakeholders, who have a substantial impact on the company's success. Businesses, as well as groups and individuals with commercial interests and affiliations, such as consumers, employees, suppliers, etc. Because stakeholders can contribute to a company's potential to earn a profit (Post et al., 2002). Enterprises should pay attention to the interests of stakeholders and consider their perspectives and activities in order to stabilize development. At that time, the firm is actively engaging in CSR projects with stakeholders, accelerating the company's growth and prosperity. At the same time, the company will be highly appreciated, the credit score will be skyrocketed, more prestige and the company image became attractive to investors. This will make it much easier for the organization to obtain long-term

financing. Therefore, we can hypothesize that CSR has a positive relationship with debt maturity structure.

2.4 Signaling Theory

Based on information asymmetry proposed by Jensen and Meckling (1976) and developed by (Myers, 1977). Businesses with a high level of social responsibility will issue their quality and prestige to the market by publishing short-term debt. Issuing short-term debt gives firms good indications of the firm's financial health (Flannery, 1986; Diamond and Rajan, 2001), and earnings volatility (Stohs and Mauer, 1996) when they examine control the financial risks and interest rates of short-term debt. In case, there is any information asymmetry gap between the borrowers and the lenders, the company will enhance the signaling implementation by providing information about the company's activities to minimize information asymmetry among stakeholders. On the other hand, companies operating in CSR can also be easier to issue short-term debt with low borrowing costs achieved through CSR. From the above, the research team found that CSR has a negative relationship with debt maturity structure.

3. OVERVIEW OF PREVIOUS RESEARCHES

The research of Masulis, R. W., & Reza, S. W. (2015) argues that focusing only on volunteering can cause adverse effects due to uncontrolled investment. In addition, Ye, K., & Zhang, R. (2011) also pointed out that the highest growth in CSR raises the cost of debt for enterprises. This results in CSR activities having a negative impact on the interest payment of the business, reducing the value of the company and increasing the risk of default. Banks and creditors frequently decrease the duration of debt to monitor and penalize management, and corporations are not permitted to issue long-term debt.

Based on Friedman's (1970) argument and shareholder theory argued that a firm's primary responsibility is to maximize shareholder wealth, CSR investments can lead to deviations from optimal investment decisions optimized to maximize value. Good managers may seek to over-invest in social responsibility for their own personal benefit (Barnea et al. Rubin, 2010) and job security (Cespa and Cestone, 2007), which may come at the expense of shareholders' capital.

The study (Benlemlih. M, 2017) found a negative relationship between CSR and debt maturity, especially for small enterprises, based on data from corporations in the United States. On a large scale, high CSR performance results from signal theory, because these companies invest in CSR with the desire to collect short-term debt in order to inform the quality of their performance to the market. Due to transaction costs, issuing short-term debt frequently can be costly for companies. Low-quality companies cannot afford these high transaction fees and would not want lenders to closely monitor and assess their credit risk. Therefore, low-quality businesses will try to avoid issuing short-term debt. However, in cases when confidential corporate information is not available, good quality companies tend to issue short-term debt for signaling purposes.

That is, the issuance of short-term debt can be perceived by the market as a positive signal about the credit risk and liquidity of the borrower (Flannery, 1986). Diamond (1991) argues that high-quality firms with low credit risk and low-quality firms with high credit risk will use short-term debt. Poor quality companies are less able to access the long-term debt market, and high-quality companies with low credit risk expect to receive positive signals. In this case, only medium-quality companies use long-term debt. Previous studies have shown that companies operating in CSR have lower credit risk ratings, as CSR-conscious companies are said to have better risk management capabilities (Feldman et al. et al., 1997) and rating agencies reflect such non-financial information in their assessments (Attig et al., 2013). This means that, based on the signaling argument, firms with high CSR will signal quality to the market by issuing short-term debt. Firms operating in CSR are also more likely to issue short-term debt with lower borrowing costs achieved through CSR (Goss and Roberts, 2011). Accordingly, we propose the first hypothesis as follows:

H1: CSR has a negative relationship with debt maturity structure

Based on research on CSR of listed companies, especially in the mining industry in Australia, (V.H. Nguyen, B. Choi and F.W. Agbola, 2019) conclude that proactive CSR activities can help companies generate revenue, attracting long-term capital sources from the debt market to reduce refinancing risks, and at the same time, there are positive signs contributing to increasing the future assessment of enterprises in terms of competitiveness, operational efficiency and attractiveness for investors.

Based on tax theory, Stohs, M. H., & Mauer, D. C. (1996) found some optimistic signals for taxes owing to the usage of long-term debt, that is, the benefit from tax shields. As the term structure of interest rates decreases, firms tend to choose long-term debt. As a result, managers play a critical role in deciding on CSR measures to extend the maturity of the company's debt. According to the stakeholder theory, firms must be transparent about their social duties to stakeholders so that they can trust the company to always protect their interests. Therefore, debt financing decisions depend on the degree of operational risk in the business and information asymmetry. Francis, J., LaFond, R., Olsson, P. M., & Schipper, K. (2004) reported that participating in liquidity management to enhance information disclosure, reduce asymmetry can help businesses access long-term debt financing.

Evaluation of the research results of Thao, N. T., & Phuong, P. H. (2017) in the context of enterprises in Vietnam. The author believes that shareholders should have a deeper insight into the direction of corporate social responsibility development for long-term benefits. When a company invests in community activities, the direct interests of its shareholders may be harmed. However, in the future, this will be an investment that brings high efficiency in business operations due to trust and relationship building with groups of common interests. This shows that corporate social responsibility is a worthwhile investment because it brings sustainable benefits to the company and stakeholders. Therefore, the following alternative hypothesis is proposed:

H2: CSR has a positive relationship with debt maturity structure

CHAPTER 3: METHODOLOGY

1. RESEARCH METHODS

The study's authors use two main approaches for estimating panel data, OLS - Ordinary Least Squares regression model, and GMM system - Generalized method of moments model to ensure that the research findings are robust and reliable. Furthermore, the OLS model is just a first step in analyzing the relationship between the independent and dependent variables. At the same time, OLS has the drawback of being unable to solve the common data table phenomenon, which occurs when a data set contains several companies over a long period. Besides that, OLS has difficulty controlling two phenomena in panel data: autocorrelation and variance.

System GMM was the main estimation method used by the research team to get the final result. Because it uses instrumental variables to handle the limitations of the OLS regression model as well as the 2-way influence between the explanatory variable and the dependent variable (Endogeneity) (Roodman, 2009). GMM also uses two standard tests, the AR(2) test and the Hansen test, to quantify the model's validity and effectiveness.

2. RESEARCH MODEL

2.1 Models

Our baseline model is used to test the effect of CSR on adjusted debt maturity. In line with Benlemlih (2017), we propose the following model:

$$DEBTMATi,t = \alpha 0 + \alpha 1csr + \alpha 2lev + \alpha 3assetmat + \alpha 4grow + \alpha 5size + \alpha 6term$$

In there:

i= 1, 2, 3,..., 220 enterprises;

t: respectively from 2014, 2015,..., 2018.

Dependent variable

DEBTMAT: Debt maturity;

Group of explanatory variables

Csr: Corporate Social Responsibility;

Lev: Firm leverage;

Assetmat: Asset maturity structure;

Grow: Growth opportunity;

Size: Company size;

Term: The term structure of interest rates.

2.2 Figures

From 2014 to 2018, data was collected from the financial statements of 220 companies registered on the Vietnamese stock exchange. Through panel data collected from Thomson Reuters, we conduct analysis of the impact on debt maturity structure while implementing social responsibility through 3 aspects: conomics-related CSR (CSR1), environment-related CSR (CSR2) and society-related CSR (CSR3).

2.3 Variables

2.3.1 CSR variable

Based on the combination of disclosure requirements of the Ministry of Finance in Circular 155 and GRI-GSSB series of standards, the three dimensions of CSR disclosure include economic, environmental and social effects of a firm's operations. The analysis is based on a list of indicators classified into three groups of aspects in accordance with the GRI standards: 1) economic responsibility (6 standards) including the economic impacts of the business on the locality and government; 2) environmental responsibility (8 standards) including corporate impacts on the environment, and 3) social responsibility (19 standards) includes business support activities to employees, citizens and local government, government, customers, and suppliers. The CSR aspect j in year t of company i is calculated as follows:

$$\mathrm{CSR_{ij}} = \frac{\sum_{k=1}^{n} \mathrm{SCORE_{jk}}}{n}$$

In there: CSRij is the sum of all scores that the firm receives for aspect j divided by total number of standards for aspect j ($0 \leq CSRij \leq 1$). n equals 6, 8, 19, representing the number of criteria for each aspect j.

2.3.2 Explanatory variables

2.3.2.1 Firm leverage

Measured as a ratio of total debt to total assets, showing the extent to which financial loans are used in activities that bring profitability to the business or equity.

According to Diamond (1991), financial leverage can increase liquidity risk for the business. According to Nghĩa, L. T. T., & Hùng, P. M. (2016), for small and medium enterprises, the liquidity ratio is low and the leverage ratio is lower. In contrast, firms with high competitiveness, strong position and growth potential, together with agency costs and low probability of bankruptcy, lead to the possibility of the highest leverage ratio. As a result, organizations with larger financial leverage have more opportunities for using long-term debt to postpone liquidity risk. Consequently, *financial leverage is said to be proportional to the time of debt maturity.*

2.3.2.2 Size

Calculated by the natural logarithm of total assets, based on which to control credit quality for debt maturity (Zheng et al., 2012). Larger and higher quality firms can get long-term debt more easily than smaller firms due to high credit quality (Diamond, 1991; Ben-Nasr et al., 2015).). In contrast, smaller firms tend to be excluded from the long-term debt market due to reduced credit quality due to higher levels of information asymmetry and agency conflicts between a company's management and the company's stockholders… According to research by (Beven & Danbolt, 2002), it can be seen that *firm size is inversely proportional to short-term debt and proportional to long-term debt.*

2.3.2.3 Assetmat

Maturities for assets owned by the business that can yield future returns are determined by the average value of current and long-term assets as they come due. The maturity of current assets is calculated as the ratio of total assets, plant, and equipment with depreciation and amortization. Myers (1977) suggested that firms should match the maturity of their liabilities with their assets. Thus, a company with asset maturities longer than the debt maturity date will result in default due to lack of cash, and vice versa, if a company has a longer maturity of liabilities than assets, lead to cash flows from its assets will cease, while its debt payments will come due (Stohs and Maurer, 1996). Therefore, long-term debt companies should have a lot of long-term assets. Thus, it can be said that *the asset term structure will influence positively the time that businesses want to mature their debt.*

2.3.2.4 Term

The effects on debt maturity of corporate interest rates are known as the term structure of interest rates, thereby reflecting the correlation between interest rates of debt instruments with different maturities but similar features risk structure, income tax, and liquidity is calculated by the difference between yields on 10-year government bonds and yields on 1-year bonds, represented by a 3-shaped curve: upward - reflects expectations of future short-term interest rate increases, horizontal or downwards - reflects insignificant and substantial declines in short-term interest rate expectations compared with theoretical expectations. Firms tend to use longer-term debt to take advantage of tax benefits assuming that the term structure of interest rates is upward sloping (Huang et al., 2016). Therefore, *the maturity of the debt will be proportional to the term structure of the interest rate.*

2.3.2.5 Grow

Growth opportunities are calculated by the current year's total assets minus the previous year. According to Lemma and Negash (2012) have shown the relationship of growth opportunities between countries and between debt maturity structure, particularly, *growth opportunity and debt maturity structure are negative correlations*. According to agency theory, companies that are indebted will cause conflict between shareholders, managers, and bondholders. This reduces the possibility of investing in positive net worth projects because they fear that the returns demanded by bondholders will lead to non-investment, so the use of short-term debt is seen as a solution to reduce conflicts.

CHAPTER 4: RESEARCH RESULTS

1. DATA DESCRIPTION

Variables	Observation	Mean	Std. Dev.	Min	Max
debtmat	961	0.360	0.335	0.000	1.000
csr1	1092	0.234	0.140	0.000	1.000
csr2	1092	0.312	0.266	0.000	1.000
csr3	1092	0.244	0.137	0.000	0.947
lev	1092	0.246	0.190	0.000	0.736
size	1092	28.184	1.286	23.330	31.991
grow	1057	1.372	0.891	0.243	9.058
assetmat	985	16.778	58.559	0.000	1230.561
term	1092	0.018	0.005	0.011	0.025

Table 4.1: Data Description

Observing the results from the table, we see that the average value of CSR1, CSR2 and CSR3 in terms of economics, environment and society, respectively, is relatively low, namely 0.234, 0.312 and 0.244. This is quite understandable given that Vietnam is a developing country with a low growth rate when compared to developed countries in particular or the world in general. As a result, most Vietnamese enterprises are small in scale and young, unable to fund and develop all elements of their operations, particularly in terms of social responsibility and corporate governance. Businesses still have limited access and face several cost limitations.

Following that, the average value of the debt term structure is around 0.360, indicating that the total long-term debt accounts for about 36% of the debt maturity structure of the Vietnamese enterprises under consideration, and at the same time, the ratio of the total long-term leverage of enterprises ranges from 0% to 100%. This ratio shows that the sample of Vietnamese businesses has a greater ratio of short-term debt

than long-term debt, and in general, the percentage of Vietnamese businesses utilizing debt as a source of capital is uneven.

2. ANALYSIS OF ORDINARY LEAST SQUARES REGRESSION TEST (OLS)

	csr1	csr2	csr3
CSR_1	-0.207*		
	(0.081)		
CSR_2		-0.055	
		(0.045)	
CSR_3			-0.194*
			(0.081)
lev	-0.007	-0.009	-0.006
	(0.067)	(0.067)	(0.067)
size	0.071***	0.066***	0.069***
	(0.010)	(0.010)	(0.010)
grow	-0.023	-0.028*	-0.023
	(0.014)	(0.014)	(0.014)
assetmat	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
term	0.353	0.210	0.220
	(2.226)	(2.287)	(2.234)
Constant	-1.610***	-1.488***	-1.542***
	(0.280)	(0.275)	(0.275)
R-squared	0.061	0.055	0.060
N	845	845	845

* p<0.05, ** p<0.01, *** p<0.001

Table 4.2: Results from OLS regression model

Through correlation coefficients, the OLS regression table shows that two separate components of CSR, economic CSR (CSR1) and society CSR (CSR3), have a negative influence on debt maturity structure which is -0.207 and -0.194 at 5% significance level, respectively. This result supports the hypothesis **H1:** CSR has a negative relationship with debt maturity structure, and is consistent with the findings of Ye, K., & Zhang, R. (2011) also showed that the higher implementation of CSR raises the cost of debt for businesses, leading to activities that have a detrimental influence on corporate

performance, interest payments, decline in firm value and a higher chance of default. At that time, banks and creditors often shorten the debt maturity, and businesses are not allowed to issue long-term debt for the purpose of monitoring managers. It is also consistent with the findings of (Barnea and Rubin, 2010) that competent managers may want to overinvest in CSR for their own personal gain, even if it may come at the expense of their own capital shareholders, as a result, investors will later limit their long-term financial contributions to these firms.

Particularly, environmental CSR (CSR2) has no such effect, as evidenced by the lack of statistical significance despite the presence of a negative correlation coefficient.

In short, testing the OLS model demonstrates that CSR1 and CSR3 are statistically significant, while CSR2 has absolutely no statistical significance, implying that only the implementation of CSR's economics and society have a negative influence on the debt maturity structure of the sample of firms. However, this is merely a "sign", assisting the research team in considering and making preliminary judgements before moving on to the next model and drawing final findings.

3. ANALYSIS OF GENERALIZED METHOD OF MOMENTS (GMM)

Although using the OLS model to find the correlation between the variables, the deviations in endogeneity still occur while testing the relationship between the debt maturity structure of the sample of listed Vietnamese enterprises and social responsibility based on three aspects related to economics, environment and society.

To overcome this problem, previous studies used instrumental variable estimation (IV estimator) to include in the model. And the problem when applying instrumental variable estimates is that sometimes it is not easy to find suitable instrumental variables because if weak instrumental variables are chosen, the IV estimate is likely to be biased (Mileve, 2007). This means that applying the IV estimator without finding an appropriate instrumental variable, the OLS regression estimation issues will remain. Therefore, from the study of (Arellano and Bond, 1991), the dynamic table data model System GMM is proposed to use.

The research team ran the System GMM regression model and the following results were obtained:

	csr1	csr2	csr3
L.debtmat	0.719***	0.811***	0.666***
	(0.069)	(0.055)	(0.075)
CSR_1	-0.349***		
	(0.078)		
CSR_2		-0.037	
		(0.038)	
CSR_3			-0.230*
			(0.097)
lev	0.291*	0.346**	0.338*
	(0.115)	(0.112)	(0.139)
size	0.022*	0.003	0.015
	(0.009)	(0.008)	(0.009)
grow	0.030*	0.021	0.024
	(0.015)	(0.014)	(0.015)
assetmat	0.001*	0.001***	0.000
	(0.000)	(0.000)	(0.000)
term	2.263	1.321	1.829
	(1.554)	(1.177)	(1.544)
Constant	-0.611**	-0.192	-0.421
	(0.219)	(0.216)	(0.223)
R-squared			
N	696	696	696
AR(2) test pvalue	>0.1	>0.1	>0.1
Sargan/Hansen test pvalue	>0.1	>0.1	>0.1
* p<0.05, ** p<0.01, *** p<0.001			

Table 4.3: Results from the System GMM regression model

The advantage of the GMM model over the instrumental variable estimator is that the GMM model is easier to select the instrumental variables because the application of exogenous variables to other time periods or taking the lags of the variables can be used as tool variables for endogenous variables at the current moment.

To control for the dynamics of the debtmat, the GMM regression model added L.debtmat as the lagged variable of the dependent variable debtmat, while the OLS regression model did not have this feature. From the GMM table, it is found that the correlation coefficient of the variable L.debtmat has a positive sign of 0.719, 0.811 and 0.666, and has a pretty strong significance level of 0.1%. This suggests that the previous

year the business had a lot of short-term debt to finance the purposes of working capital, the next year, there would normally not be much change in the firm's debt maturity structure. The coefficient of variable L.debtmat was statistically significant, indicating the appropriateness of using the dynamic model in our research.

At the same time, the GMM regression results show that the different components of CSR, namely economics (CSR1) and society (CSR3), have a negative impact on the debt maturity structure of the sample of Vietnamese enterprises with the significance levels of 0.1% and 5%, respectively, through two negative correlation coefficients -0.349 and -0.230. Particularly, environmental CSR (CSR2) does not have this impact, as evidenced by the lack of a statistically significant level despite a negative association with the dependent variable. This estimation result is consistent with the results of the OLS panel data model, that the effect of CSR, specifically CSR1 (economics) and CSR3 (society) on the debt maturity structure of a sample of listed firms in Vietnam is to lower the dependent variable debtmat.

It is worth noting that the research team ran the results of Hansen's test on the validity of the model and the Abond test (AR(2) - autocorrelation test of order 2). The Hansen test shows that the p value is greater than 0.1, indicating that the initial hypothesis is rejected. Therefore, the model is valid, effective estimation (Overidentification). Furthermore, AR(2) test gives p value greater than 0.1, that is, the original hypothesis about the absence of 2-order series correlation is rejected. As a consequence, all the results in the GMM regression model are significant, the correlation coefficients are reasonable for interpreting the statistical results.

Conclusion: We have taken the approach in different ways, but both the findings obtained from the OLS and GMM models support the hypothesis H1 given by the reseach team: The Social responsibility performance is negatively related to the debt maturity structure of Vietnamese listed companies, which agrees with offered theories such as: shareholder theory, agency theory and signal theory. This result is also similar the author's earlier research such as Ye, K., & Zhang, R. (2011), (Barnea and Rubin, 2010), (Benlemlih, M., 2017) when studying the American business environment…And disagree with the view that aggressively implementing CSR will extend the debt maturity of enterprises through the studies of (V.H. Nguyen, B. Choi and F.W. Agbola, 2019), (Thao, N. T., & Phuong, P. H, 2017),…

CHAPTER 5: CONCLUSION AND RECOMMENDATION

1. CONCLUSION

Firstly, the findings support three theories: shareholder theory, agency theory, and signal theory, and are compatible with hypothesis **H1:** "CSR has a negative relationship with debt maturity structure." Also, we agree with some of the findings of previous research.

Second, we discover that CSR is a vital component in enhancing business performance, creating chances to conveniently access debts, expanding company's production and commercial operations, reducing information asymmetry, and increasing the prestige of enterprises in the market.

Third, based on the findings, we believe that CSR will facilitate debt financing and will be useful in establishing corporate policies to better implement social responsibility in the future, ensuring long-term development for businesses and society.

2. RECOMMENDATION

Social responsibility is a significant aspect in increasing economic performance, offering options for debt financing, reducing information asymmetries, and ensuring that businesses and companies maintain the highest level of market reputation. More businesses are devising strategies to engage in activities that exhibit social responsibility in order to benefit themselves and others. The group's analysis in the essay shows that CSR has a detrimental impact on firm debt maturity structure in three aspects: economic, environmental, and social. Lenders are willing to supply organizations with greater CSR with shorter-term loans. Because financial stability helps greatly to business productivity, attracting debt-based source capital provides an assurance of financial supply as well as appropriate risk management. CSR facilitates short-term debt financing, which can assist businesses in implementing CSR activities in the future. However, the issue of corporate social responsibility (CSR) in Vietnam is still new, has received little attention, and has a number of difficulties. As a result, businesses and politicians should continue to raise CSR awareness through the initiatives listed below:

(1) Paying close attention to CSR implementation

In the long term, every company should research and implement green solutions as well as create energy-saving technology. If you exhibit environmental responsibility, consumers and stakeholders will have more trust in your company.

Businesses can assist the community by donating to charity, contributing to a community fund, or establishing vocational training programs in the neighborhood. Furthermore, companies can work with other financial partners to assist and share the cost of financing activities that benefit the community, resulting in the best possible outcomes.

Companies must detect and enforce social responsibility in the workplace based on employee preferences. At the same time, businesses should enable employees to take the lead in CSR by allowing them to make their own proposals and structure them so that employees recognize themselves as a critical component of the business.

(2) Raising customer awareness of corporate social responsibility

The authors' findings also show an increase in consumer interest, which has a significant impact on their decisions. As a result, capturing consumer judgment in order to conduct appropriate rules to help build business reputation, trust, and customer loyalty. Furthermore, using corporate social responsibility as a management tool is critical, and the small business industry, which is at the forefront of CSR implementation, requires a unique marketing and service approach.

(3) Increase leaders' social responsibility awareness.

Managers and executives should be assisted in learning about the nature and implementation of CSR through seminars and workshops. Access to training programs will assist leaders in grasping the fundamentals and core of CSR, as well as understanding the substance of CSR and being inspired to put it into practice. CSR implementation will fail if the leaders do not understand CSR and do not actively promote CSR activities, so CSR must emanate from the leader.

(4) Creating social relationships with suppliers

There are a lot of tasks a firm can do right immediately with suppliers, such as establishing clear, precise product and service specifications, and guaranteeing effective communication. We also need to evaluate and implement operational measures such as guaranteeing to pay the contract on time based on an analysis of the supplier's demands before beginning to arrange production, rationalizing production and inventories,

avoiding budget stagnation, and preventing supplier debt deferral. Furthermore, businesses can contribute to increasing the quality of raw materials by providing financial, technological, scientific, and technical support for production, packaging, management, and so on.

LIST OF TABLES

Table 4.1: Data Description

Table 4.2: Results from OLS regression model

Table 4.3: Results from the System GMM regression model

LIST OF ABBREVIATIONS

1. CSR	Corporate Social Responsibility
2. etc.	and other similar things
3. OLS	Ordinary Least Squares regression
4. GMM	Generalized method of moments

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